Aravinth K

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Summary

Motivated and detail-oriented B.E. Electronics and Communication Engineering graduate passionate about Artificial Intelligence and Machine Learning. Skilled in developing data-driven models and real-world software solutions using Python and ML algorithms, with hands-on experience in building computer vision systems, predictive models, and IoT-integrated analytics solutions.

Education

B.E. Electronics and Communication Engineering 2020 – 2024
K.S.R. Institute for Engineering and Technology, Namakkal
CGPA: 7.919

Technical Skills

- **Programming Languages:** Python
- **Machine Learning:** Classification, Regression, Feature Engineering
- **Deep Learning:** CNNs, Image Classification, Transfer Learning
- **Frameworks:** Scikit-learn, TensorFlow, Keras
- **Computer Vision:** Image Preprocessing, CNN-based Image Classification
- **Tools / Platforms:** Jupyter Notebook, Google Colab
- **Other Skills:** IoT Integration, Sensor Data Processing

Projects

Sleep Disorder Prediction – Machine Learning

- Developed a complete machine learning pipeline to classify sleep disorders using lifestyle and health-related data.Performed data preprocessing including outlier detection, missing value handling, encoding, and normalization. Trained and evaluated Random Forest, and Logistic Regression models, achieving best performance with Random Forest.

Vegetable Image Classification – Deep Learning

- Built a CNN-based image classification model to identify multiple vegetable categories. Applied image preprocessing techniques such as resizing, normalization, and data augmentation. Achieved 92% accuracy using a fine-tuned multi-layer CNN with TensorFlow and Keras. Evaluated model readiness using training and validation performance metrics.

IoT-Based Health Monitoring System

- Designed an IoT-driven system for real-time physiological data collection and monitoring. Developed cloud-connected workflows with visualization, alerting, and anomaly detection. Implemented instant alerts using Telegram for abnormal sensor readings. Created dashboards for real-time tracking and historical trend analysis.

Certifications

- Programming Essentials in Python – Cisco Networking Academy
- Skill Employment Training Certificate
- Career Development Certificate
- AWS Educate: Machine Learning Foundations
- AWS Educate: Introduction to Generative AI
- AWS Educate: Getting Started with Databases
- AI Programmer Training – Completed

Strengths

- Quick learner, disciplined, and passionate about continuous learning.

Achievements

- District Winner – Silambam